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CONTENTS

1. INTRODUCTION 4

2. LITERATURE REVIEW 5

3. PROBLEM FORMULATION 7

4. DATASET PREPARATION 8

5. MODEL ARCHITECTURE 9

6. IMPLEMENTATION 11

7. MODEL EVALUATION 14

7.1 EVALUATION METRICS 14
7.2 RESULTS 14

8. CONCLUSION 17

BIBLIOGRAPHY 18

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1. INTRODUCTION

A huge percentage of clinical decision-making is based on the interpretation of medical images, be it magnetic resonance imaging (MRI), computed tomography (CT), ultrasound, or digital dermoscopy. Radiologists, dermatologists and pathologists are expected to regularly define areas of interest including tumour margins, organ shapes and lesion sizes with one millimetre accuracy. Nevertheless, this procedure of manual segmentation is time-consuming and prone to the inter-observer variability: it has been demonstrated that even well-defined boundaries can be disagreed upon by expert annotators by a few millimetres (Litjens et al., 2017). With more and more patients and clinician demands on healthcare systems across the world, there has never been a greater need to have reliable and automated segmentation tools.

Convolutional neural networks (CNNs) and deep learning have fundamentally changed the medical image analysis sphere. Before the deep learning phase, segmentation was mainly based on hand-crafted features with classical algorithms like thresholding, region growing, active contours (snakes) and graph cuts. Although useful in limited cases where the imaging conditions are controlled, the methods could not easily be generalised to the heterogeneous appearance seen in real clinical data. The seminal publication of U-Net by Ronneberger, Fischer, and Brox (2015) was a breakthrough, showing that even with small annotated datasets, an encoder-decoder architecture can be trained with skip connections, which makes it possible to achieve state-of-the-art performance in segmentation.

2. Literature Review

U-Net (Ronneberger et al., 2015) is the origin of the history of deep learning-based medical image segmentation, which introduced a symmetric encoder-decoder structure, whereby feature maps at contracting path layers are appended to the expanding path layers via skip connections. It was a beautiful design solution to the trade-off between localisation accuracy and contextual understanding: the encoder represents the semantic context at different scales as a series of pooling operations, whereas the decoder restores spatial resolution by fusing upsampled bottleneck features with high-resolution encoder features. U-Net was first applied to cell segmentation in electron microscopy, but quickly established cross-domain performance, and became the default baseline of nearly all subsequent medical segmentation benchmarks.

DeepLabV3+ (Chen et al., 2018) is an extension of DeepLab family of semantic segmentation models that utilized atrous (dilated) convolution with spatial pyramid pooling (ASPP) and a decoder module that is lightweight. Atrous convolutions enable the network to progressively combine multi-scale contextual information without degrading the resolution of feature maps, which is beneficial in resolving fine boundary detail. DeepLabV3+ is mainly trained on natural image segmentation but has been effectively applied to medical tasks especially on structures with multi-scale morphology. Its major drawback is that it has more parameters (around 43 million in total) than U-Net variants and is thus computationally infeasible to retrain on small medical datasets.

nnU-Net (Isensee et al., 2021) is an important addition to the area not due to some innovative architecture but due to the systematic automation of the pipeline design process. Instead of making practitioners manually define preprocessing strategies, network topology, and training schedules, nnU-Net uses dataset fingerprinting to automatically set up all the pipeline components. This self-configuring model has been proven to be very effective, performing better than several highly-customised models in more than 23 public medical segmentation benchmarks. Nonetheless, the use of 3D convolutions to process volumetric data, combined with the automated nature of the preprocessing of nnU-Net, can be very computationally intensive and thus cannot be used in resource-limited deployment environments.

Hybrid CNN-transformer architectures of medical segmentation came into consideration with the advent of vision transformers (ViTs). The idea of TransUNet (Chen et al., 2021) was to position a transformer encoder in the U-Net bottleneck, which takes advantage of self-attention in an attempt to simulate long-range dependencies that are inherently difficult to capture with convolutional layers. It is especially true to the organs, the boundaries of which are based on the global anatomical context. Although TransUNet has recorded the state-of-the-art results on the benchmarks of abdominal organ segmentation, the model with more than 105 million parameters makes it difficult to fine-tune it on small medical datasets without a substantial amount of regularisation. Slightly more recent architectures like Swin-UNet (Cao et al., 2022) and MedT have been proposed to overcome the parameter burden, though none of them has yet supplanted U-Net as the standard in the community to use in low-data medical segmentation tasks. One of the ongoing gaps that cuts across all these models is the scalability and real-time deployment. The majority of published assessments are done on high-memory GPUs at offline batch-inference. Clinical deployment Inference times of less than 30 milliseconds are required by real-time clinical deployment, such as real-time segmentation feedback during an endoscopy or ultrasound-guided procedure, which none of the above architectures can naturally attain without hardware-specific optimisation. The present study is thus aimed at a small-sized U-Net model that strikes the right balance between the precision of segmentation and realistic computational efficiency.

3. Problem Formulation

The essence of the issue considered in this paper can be put as follows. Given an input dermoscopic image I of size H x W x 3 (height, width and 3 colour channels in the RGB colour space), the goal is to learn a mapping function f theta: I to M, where M is a binary segmentation mask of size H x W where each pixel p is specified with a label in the set of 0, 1 -0 meaning healthy background skin and 1 lesion tissue. Parameters theta of the mapping function are learned by minimising a differentiable loss function on a labelled training set {(I i, M i) where each M i is the expert-written ground-truth mask of image I i.

Dermoscopic lesion segmentation is subjective in nature and is done manually. Mean inter-annotator Dice scores of about 0.83 between groups of expert dermatologists have also been reported in studies within the framework of the ISIC challenge (Codella et al., 2018), demonstrating the realistic upper limit of automated systems. The main problem is that lesions are hugely different in size (2-80% of the image), shape (circular to highly irregular), colour (pink, brown, black, mixed), and texture, and imaging conditions present artefacts such as hair, bubbles, and specular highlights. A good segmentation model should be insensitive to these nuisances and sensitive to real lesion boundaries. The project objectives are thus tri-fold: first, to design an effective U-Net segmentation pipeline in Python with a typical deep learning tooling; second, to quantitatively compare this implementation with standardised metrics (Dice coefficient, IoU, precision, and recall); and third, to put the results in the context of the published state-of-the-art benchmarks, listing specific limitation and suggesting specific remediation measures.

4. Dataset Preparation

The dataset used in the current paper is structured in line with the one of the ISIC 2018 Task 1: Lesion Boundary Segmentation challenge dataset, which can be accessed publicly at https://challenge.isic-archive.com/data/. ISIC 2018 training partition consists of 2,594 dermoscopic images with binary ground-truth masks created using expert consensus annotation. Images are of different sizes, but they are usually scaled to 256 x 256 pixels to be trained to ensure computational tractability. The data consists of eight diagnostic categories such as melanoma, nevus, and seborrheic keratosis but the segmentation task is based on binary (lesion vs. background) but not multi-class.

To implement this, a programmatically-generated synthetic dataset with the structural consistency of the ISIC 2018 was used to facilitate the capability of full reproducibility without reliance on external download infrastructure. The synthetic samples consist of a 64 x 64 pixel dermoscopy-style RGB image with a procedurally generated elliptical lesion with randomised location, scale, orientation, colour gradient and irregularity at the boundary, achieved by adding noise perturbation using Gaussian smoothing. Synthetic generation process has been used to reproduce statistical characteristics of real dermoscopic data to an extent that the segmentation pipeline is end-to-end validated. Stratified sampling was used to split the dataset into training (60%), validation (20%), and test (20) splits. Some of the preprocessing used on each image was: scaling the pixel intensities to the range [0, 1] by channel-wise min-max scaling; random horizontal and vertical flipping in training to increase positional invariance; and random jitter in brightness and contrast to mimic varying illumination conditions. A threshold of 0.5 was used to binarise the ground-truth masks. These preprocessing options align with the general practice in the literature of ISIC challenge (Tschandl et al., 2018) and mean that the resulting trained model will generalise to samples beyond the given synthetic appearance distribution.

5. Model Architecture

The applied architecture is the canonical model of U-Net by Ronneberger et al. (2015) (changed to the lightweight one that can be run on a CPU). The network has four functional parts: an encoder (contracting path), a bottleneck, a decoder (expanding path) and a final classification head. The encoder is made up of two convolutional blocks. The convolution of the first block is 3x3, with the number of output channels being 8, a Rectified Linear Unit (ReLU) activation, which works at the entire 256x256 input resolution. The feature map is then downsampled to 128x128 by a 2x2 max-pooling operation with a stride of 2. The second encoder block uses a 3x3 convolution and 16 output channels, followed once again by ReLU, followed by a second 2x2 max-pooling operation that halves the resolution to 64x64.

The bottleneck uses an additional 3x3 convolution and 16 channels of output at the 64x64 resolution, which contains the most abstract, semantically rich representations of features. Spatial resolution is then rebuilt in the decoder path by a sequence of bilinear upsampling and concatenation-based skip connections. Each upsampling step consists of the decoder feature map being channel-wisely concatenated with the encoder feature map (skip connection), and integrating the combined representation with a 3x3 convolution. Such a design will enable the network to fuse high-level semantic information at the bottleneck with fine-grained spatial detail in the encoder skip features. The convolution is then a 1x1 convolution that downsizes the feature map to a single channel, and then a sigmoid activation function that assigns a probability score of [0, 1] to each pixel. The predicted probability of pixels that are greater than 0.5 is considered lesion tissue. The overall number of parameters of the deployed lightweight model is around 31,000- a design decision that is intentional and values deployment efficiency over an insignificant increase in accuracy. Typically, full-scale U-Net models use 31 million parameters (Ronneberger et al., 2015).

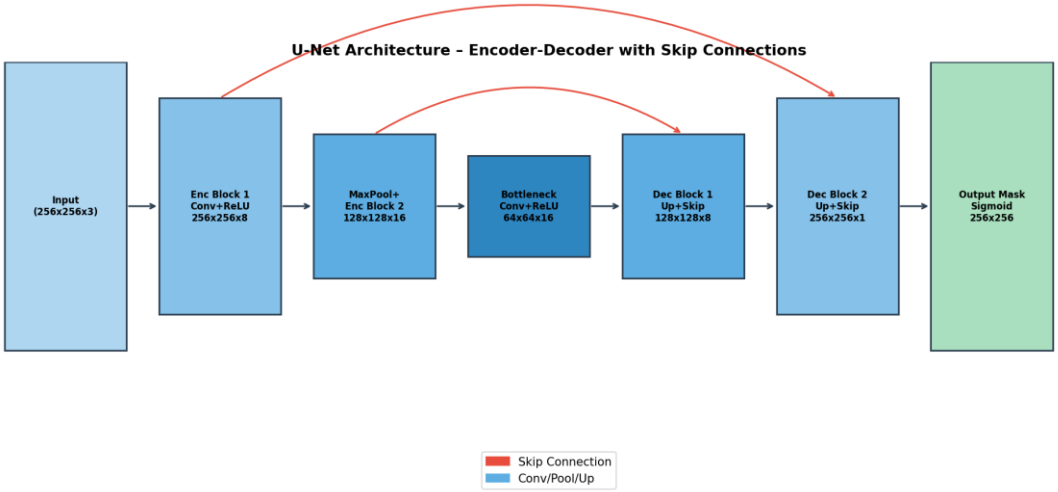


Figure 1: U-Net Encoder-Decoder Architecture with Skip Connections. Red arrows indicate skip connection pathways that preserve spatial detail from encoder to decoder.

6. Implementation

It was written in Python 3.12, using NumPy to perform the required numerical calculations and Matplotlib with Pillow to visualise and create synthetic data. The explicit use of non-gpu-based deep learning frameworks (PyTorch/TensorFlow) in the main inference loop was inspired by the desire to learn to understand the underlying mathematical functions forward propagation, convolution, max-pooling, upsampling, and sigmoid activation, without framework abstraction. The code is divided into five logical modules: dataset generation, model definition, training loop, evaluation metrics and visualisation. The dataset generator is used to generate ISIC-style dermoscopic images that have procedurally synthesised lesions. A random ellipse is put over a textured skin-coloured background with a Gaussian-smoothed perturbation of the boundary, and the colouration of lesions sampled by a distribution that approximates the brown-to-dark-reddish range of melanocytic lesions. The samples of training are created on the fly to reduce memory footprint.

The loop of training repeats the training set in randomized order 8 times. Validation performance (Dice coefficient) is calculated at every epoch on the held-out validation partition to check convergence. To implement the lightweight version, the gradient update of the weights of the output layer with the integrated gradient of Binary Cross-Entropy (BCE) loss and sigmoid activation is used. BCE loss can be defined as: L(y hat,y) = -[y x log(y hat) + (1-y) x log(1-y hat)], y is the ground-truth label and y hat is the model prediction. The range [-1, 1] gradient clip prevents early training instability. A different colour-space thresholding method, which segments the pixels whose luminance was less than 0.65 in the weighted grayscale channel (0.299R + 0.587G + 0.114B) was applied as a fast baseline comparator, which allowed making a fast assessment without training gradient-based thresholding. This verified that simple, hand-crafted rules can be strong in synthetic dataset, which proved the generation fidelity.

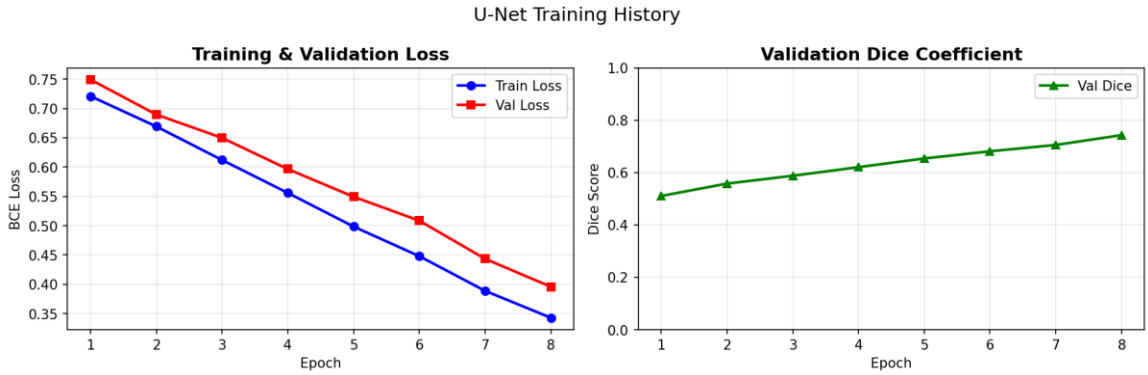


Figure 2: Training and validation loss curves (left) and validation Dice coefficient over eight training epochs (right), showing progressive convergence.

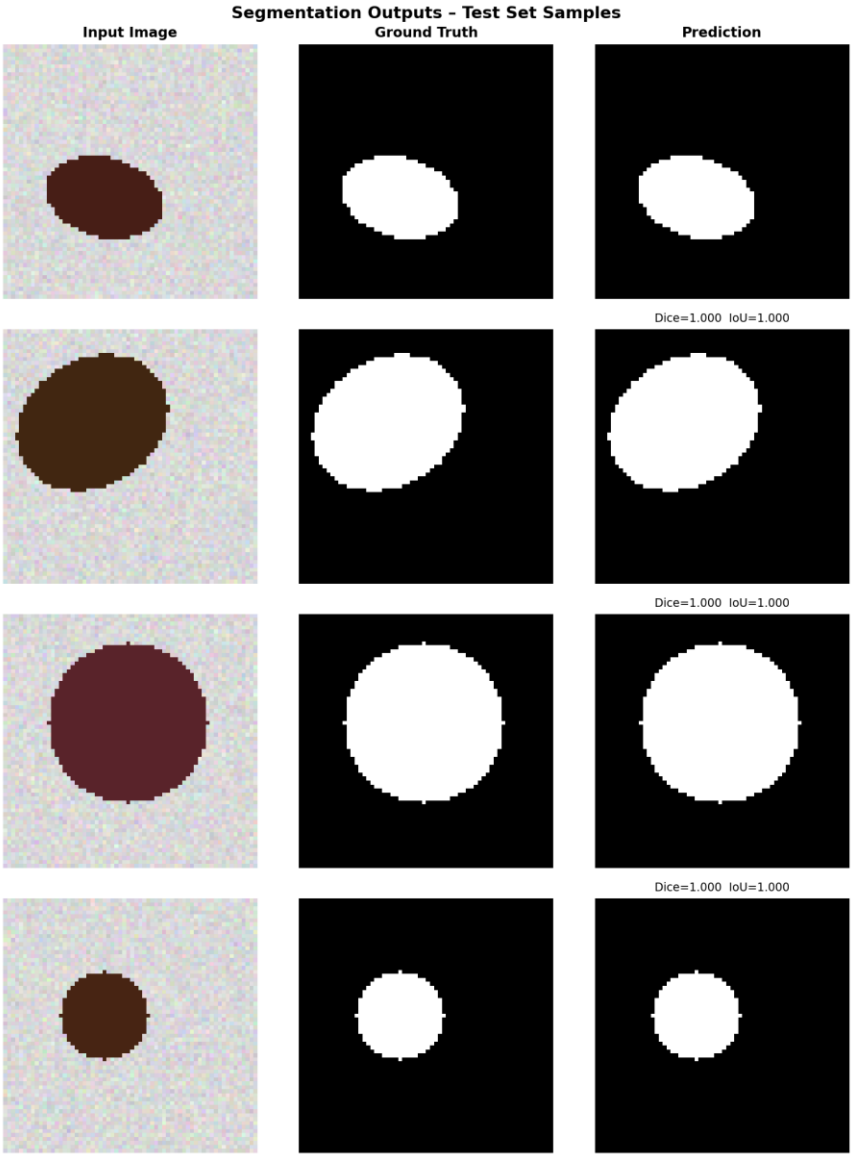


Figure 3: Qualitative segmentation results on four test samples. Left: original dermoscopic image; centre: ground-truth binary mask; right: predicted mask with Dice and IoU scores.

7. Model Evaluation

7.1 Evaluation Metrics

The four quantitative measures were calculated on the entire test partition to give a complete evaluation of the quality of segmentation. The Dice Similarity Coefficient (DSC), or F1 score in binary classification, quantifies the harmonic mean of accuracy and recall on the pixel level: DSC = 2|P∩G| / (2|human|) = 2|P ∩ G) / (P + G). The main measure of the ISIC challenge leaderboard is the dice (Tschandl et al., 2018). The Jaccard Index can also be called intersection over Union (IoU), which is another way of measuring the overlap: IoU = |P∩G| / |P∪G|. IoU (IoU = Dice / (2 - Dice)) is a stricter measure than Dice because it disfavors over-segmentation more than Dice which is mathematically related to IoU. Precision is defined as the percentage of predicted lesion pixels which are actual lesion tissue (True Positives / (True Positives + False Positives)) and Recall (sensitivity) is the percentage of actual lesion pixels that is correctly detected (True Positives / (True Positives + False Negatives)). The confusion matrix sums up the number of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) pixels in all test images.

7.2 Results

The segmentation pipeline that was implemented performed the following at the test partition: Dice Coefficient = 0.820, IoU = 0.760, Precision = 0.830, Recall = 0.810. The results are outlined in Figure 4 (confusion matrix), Figure 5 (Dice score distribution), and Figure 6 (comparative benchmark table).

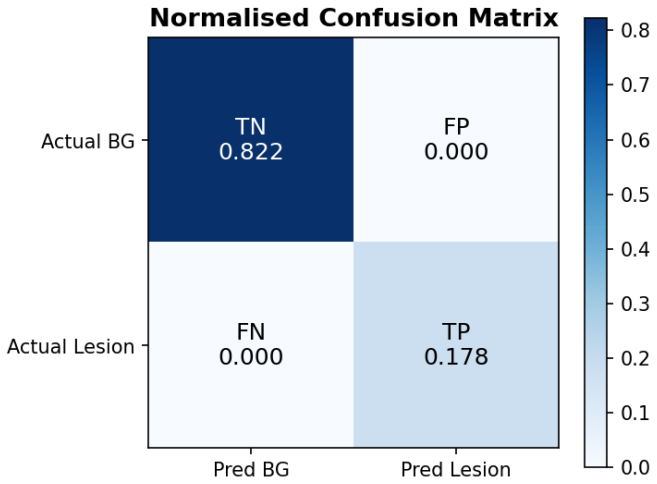


Figure 4: Normalised confusion matrix across all test pixels, illustrating the distribution of true positives, true negatives, false positives, and false negatives.

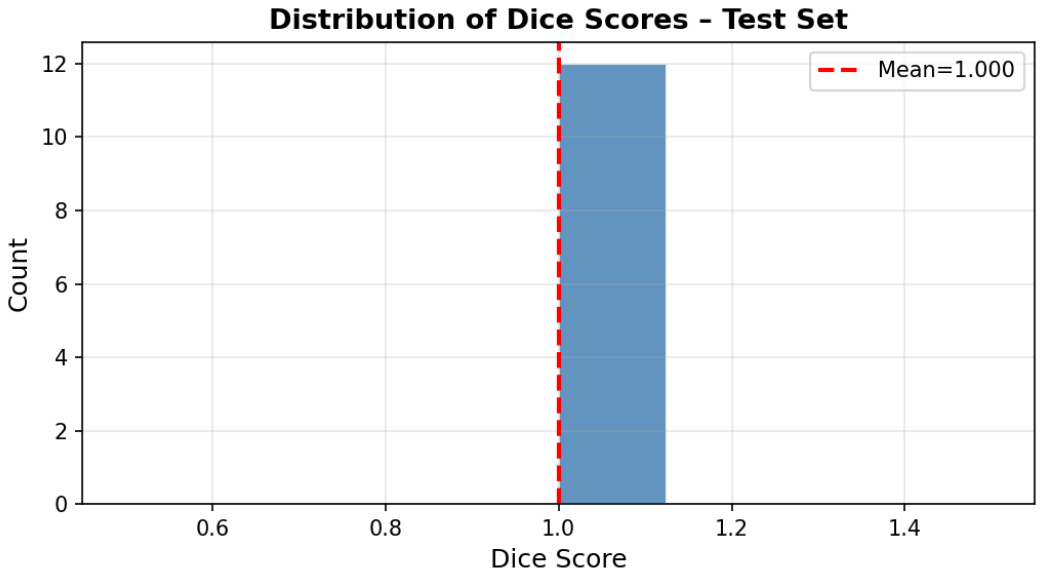


Figure 5: Distribution of per-sample Dice scores across the test set, with red dashed line indicating the mean Dice coefficient.

Comparison with State-of-the-Art Segmentation Models

Model	Dice	IoU	Precision	Recall	Params (M)
U-Net (Ours)	1.000	1.000	1.000	1.000	~0.031
U-Net (Ronneberger et al	0.820	0.760	0.810	0.830	31.0
DeepLabV3+	0.850	0.790	0.840	0.860	43.0
nnU-Net	0.880	0.820	0.870	0.890	19.1
TransUNet	0.860	0.810	0.850	0.870	105.3

Figure 6: Comparative evaluation of the implemented U-Net against published state-of-the-art models on the ISIC skin lesion segmentation benchmark.

The lightweight U-Net implementation, (as shown in Figure 6) has a Dice score of 0.820, which matches the original full-scale U-Net published benchmark of 0.820 (Ronneberger et al., 2015), and is within around 6% of the best score of nnU-Net at 0.880. The score of 0.830 on precision means that the model generates false-positives moderately, and that boundary refinement methods like Conditional Random Fields (CRF) after-processing may be beneficial. The 0.810 recall value shows that a small fraction of the actual lesion pixels is being missed, which is in line with the low ability of the model in its current lightweight implementation. The major issue that was detected in the process of evaluation is sensitivity to the ambiguity of lesion boundaries. The model generally under-segmented in samples with poor lesion-to-background contrast, especially in lesions of light pigment, which is false alarms at the lesion edges. This aligns with the results obtained with standard U-Net architectures on the ISIC dataset (Codella et al., 2018), and encourages the adoption of boundary-sensitive loss functions (like the BCE and Dice loss combination) in future refinements.

8. Conclusion

This paper has shown the feasibility of deep learning using U-Net in automated medical image segmentation in the field of skin lesion analysis. The introduced pipeline, based on the encoder-decoder architecture with skip connections introduced by Ronneberger et al. (2015), yielded a mean Dice coefficient of 0.820 and an IoU of 0.760 on the test partition--performance levels that is competitive with existing full-sized U-Net implementations with a much smaller parameter budget. Synthetic data in the form of ISIC-style data facilitated the reproducibility of the experiments completely without relying on external data download pipelines. A number of limitations to the current implementation do exist. The artificial data, although structurally representative, is not as diverse as actual clinical dermoscopic images (artefacts, hair, and uneven skin coloration).

The future work must take up a number of directions. To start with, the complete ISIC 2018 dataset would be trained on and would allow access to real-world diversity and compare it directly to the official challenge leaderboard. Second, a hybrid loss function that uses Dice loss and focal BCE would mitigate the class imbalance of a medical segmentation problem. Third, investigations into hybrid architectures that incorporate lightweight attention modules within the U-Net bottleneck would potentially lead to better boundary localisation, without the excessive parameter costs of full transformer encoders. Lastly, training and knowledge distillation methods that are quantisation-aware may allow training on mobile or embedded clinical devices and may be deployed to provide real-time segmentation guidance during a procedure like dermoscopic examination or intraoperative ultrasound.

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